

Ungulate calls: transferable context prediction with a confounding warning

The source supplies 3181 author-extracted calls from 168 animals across seven ungulate species. Source Positive/Negative labels describe eliciting behavioral contexts; they are not direct measurements of subjective emotional state. Whole animals are held out within species: 134 development animals and 34 reserved animals, yielding 475 reserved calls. Calls are weighted equally within each animal and animals equally overall.

Selected equation

The selected compact model uses species, call duration D in seconds and waveform root-mean-square amplitude R relative to digital full scale:

$$\Pr(\text{positive context}) = \sigma(\alpha_{\text{species}} - 0.69376361\ln R - 1.419715\ln(D/1\text{ s})).$$

Species intercepts are the intercept plus the corresponding indicator coefficient, with Cow as reference; the exact ordered values are in rules.json and EQUATIONS.md. The waveform amplitude is **not calibrated sound pressure**. Negative duration/amplitude coefficients characterize this dataset's recording/context structure; they are not universal emotional or physiological effects.

Five adaptive attempts compared arousal-related spectral/temporal summaries, tonality, envelope modulation, a recording-amplitude challenge and an amplitude-free spectrotemporal synthesis. The simpler duration/amplitude candidate was selected within the prespecified 1% development-error tolerance. Filename context words, animal IDs and labels are excluded from predictors. Known species is available to every comparator.

Frozen results

Model	Development group error	Reserved group error
baseline_constant	0.2055007	0.2092979
baseline_species	0.1973187	0.2059766
baseline_flexible	0.187627	0.2016159
attempt_001_arousal	0.1833687	0.1900535
attempt_002_tonality	0.1982426	0.2013305
attempt_003_modulation	0.1828181	0.1897613
attempt_004_recording	0.1821831	0.1902146
attempt_005_spectrotemporal	0.1830023	0.1891998

Selected before confirmation: **attempt_004_recording**. Primary error is animal/person/group-balanced Brier probability squared. All candidates are shown to preserve unfavorable results; a lower retrospective score does not change the selected reference.

Reserved animal-balanced Brier is 0.190215 for the selected equation, versus 0.209298 constant prevalence, 0.205977 species prevalence and 0.201616 flexible acoustics. Event precision/recall/F1 at a declared 0.5 threshold, log loss and calibration bins are retained separately; that threshold is an equal-cost diagnostic, not a welfare decision boundary. The numerical transfer supports a reproducible context descriptor, but the ability of uncalibrated RMS and duration to match richer acoustics is a warning: call type, eliciting context, species, source study and recorder can remain confounded after animal holdout. The source already demonstrates machine learning of valence, so neither a new emotion law nor deployed welfare impact is established.

Source labels WildBoar and WildBoar are linked for the ten overlapping animal IDs, preventing an otherwise hidden split leak.

Validation and reproducibility

After the complete author-extracted call. Predictors only waveform summaries and known species. File context words, animal ID, call label and source reference are never predictors. No held-animal labeled calibration. Fixed waveform summaries; any coefficients, centers or species normalization fitted only on training animals.

Seven ungulate species, source behavioral contexts and recording studies. Labels are experimental context-based valence annotations, not independently verified subjective emotion; held animals do not remove study/recording confounding. All outcomes are now exposed to later agents. Claims of new validation require fresh data or an explicitly retrospective designation. Source study and processing are disclosed in SOURCE_AND_UNITS and PRIOR_ART; this is computational confirmation, not independent experimental replication.

python run.py verifies the allowlisted package and reproduces all saved predictions. rules.json contains exact coefficients and all learned transforms; EQUATIONS.md names each feature and equation. The adjacent native adapter reconstructs source-hash-verified observations and predictors; task_spec.json declares input, timing, calibration and grouping budgets. Future proposals may use different equations under the same contract, or seek review of a genuinely different measured task.

Source: Animal-group transfer of vocal context labels; primary source/prior art. License recorded by source: CC BY4.0. Literature and semantics audited 2 October 2026.